

VirtualWorks Lab Internship

Task 3 - Causality Assessment of Drug Reactions

About this submission

This task uses three simulated adverse drug reaction cases to practise causality assessment. No patient dataset was provided with the task, so the cases were created only for internship learning. They do not represent real patients or real safety reports.

Assessment approach

For each case, I looked at the timing of the reaction, whether the event is already known with the suspected medicine, what happened after stopping the medicine (dechallenge), whether the event returned after re-exposure (rechallenge), possible alternative causes, and relevant patient factors. I then used the WHO-UMC categories as the main final classification and the Naranjo algorithm as a supporting structured check.

Naranjo score guide: 9 or more = Definite; 5-8 = Probable; 1-4 = Possible; 0 = Doubtful.

Case 1 - CA-001

Patient	MK 28 years Female 58 kg
Indication	Acute musculoskeletal pain
Suspected medicine	Ibuprofen 400 mg Oral TID after food
Therapy started	04 Aug 2026
Adverse reaction	Generalized urticaria with intense itching
Time to onset	About 2 hours after the second dose
Dechallenge	Positive - ibuprofen was stopped and the rash resolved within 24 hours after antihistamine treatment.
Rechallenge	Positive - accidental re-exposure 3 weeks later was followed by similar urticaria within about 1 hour.
Alternative causes	No new food, infection, cosmetic product, or other medicine was identified around either episode.
Relevant history	Patient recalled a similar rash after ibuprofen approximately 2 years earlier.
Outcome	Recovered
Naranjo result	9 (Definite)
WHO-UMC assessment	Certain

Why I classified it this way

The reaction followed ibuprofen on more than one exposure, improved after withdrawal, reappeared after re-exposure, and no convincing alternative cause was identified.

Case 2 - CA-002

Patient	RS 56 years Male 74 kg
Indication	Hypertension
Suspected medicine	Lisinopril 10 mg Oral Once daily
Therapy started	10 Jul 2026
Adverse reaction	Persistent dry cough
Time to onset	Approximately 12 days after treatment was started
Dechallenge	Positive - cough gradually improved and resolved within 8 days after lisinopril was discontinued.
Rechallenge	Not done.
Alternative causes	No fever, respiratory infection, smoking history, or other obvious cause was identified.
Relevant history	No previous ACE inhibitor exposure documented.
Outcome	Recovered
Naranjo result	6 (Probable)

WHO-UMC assessment	Probable / Likely
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Why I classified it this way

The timing was reasonable, cough is a recognized reaction to ACE inhibitors, no strong alternative explanation was present, and the event improved after withdrawal. Rechallenge was not required.

Case 3 - CA-003

Patient	AP 42 years Female 67 kg
Indication	Type 2 diabetes mellitus
Suspected medicine	Metformin 500 mg Oral Twice daily
Therapy started	19 Aug 2026
Adverse reaction	Diarrhea and abdominal discomfort
Time to onset	One day after treatment was started
Dechallenge	Unclear - symptoms improved over the next 2 days, but the patient also received oral rehydration and dietary advice.
Rechallenge	Not done.
Alternative causes	A family member had similar gastrointestinal symptoms during the same week, so an infectious gastroenteritis could
Relevant history	No relevant previous metformin exposure.
Outcome	Recovered
Naranjo result	2 (Possible)
WHO-UMC assessment	Possible

Why I classified it this way

The event occurred after metformin exposure and is pharmacologically plausible, but another reasonable cause existed and the effect of drug withdrawal could not be clearly separated from natural recovery/supportive treatment.

What I learned

The main point I took from this task is that an event happening after a medicine is started does not automatically prove causality. A stronger assessment comes from putting the timing together with dechallenge, rechallenge, known pharmacology, patient history and other possible explanations. The difference between Certain, Probable and Possible is mainly the strength and completeness of that evidence.

Disclaimer

All patient initials, dates, clinical events and medication scenarios in this submission are fictional and were created only for pharmacovigilance training.